Gρ Hydroponics

Mechanical Engineering
UNIVERSITY OF COLORADO BOULDER

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1. Why Hydroponics?

- Year-round growth
- Reduced plant care
- Reduced Water Use

2. Goals

- Create a small modular system to enable users to customize the size to fit their needs
- Provide clear signals to the user of the status of the greenery
- Have a clean design adaptable to many environments

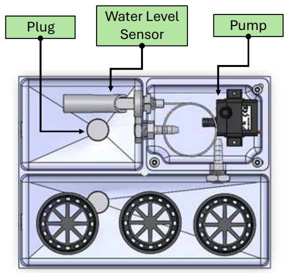
3. Requirements

- Single Unit Footprint: 10”x10”x15”
- Running Noise: ≤40 dB @ 3 ft.
- Power Input: 120V 60Hz AC
- Modular from 2+ units
- Minimize user input to power and light on/off

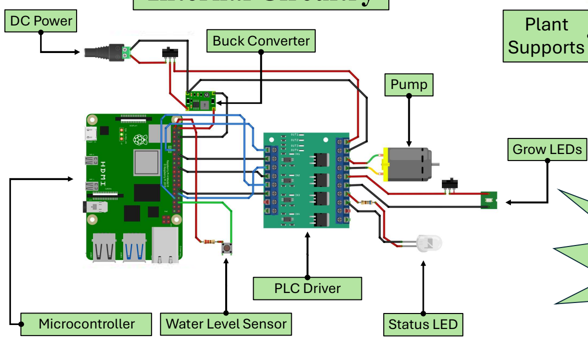
4. Manufacturing

- Followed design for manufacturability philosophies
- Milled polycarbonate paneling for high tolerance and strength
- The basin, designed for injection molding, was 3D printed in PETG for fast turnaround and low cost
- Utilizing an 80/20 frame reduced machining time, assembly time, and cost

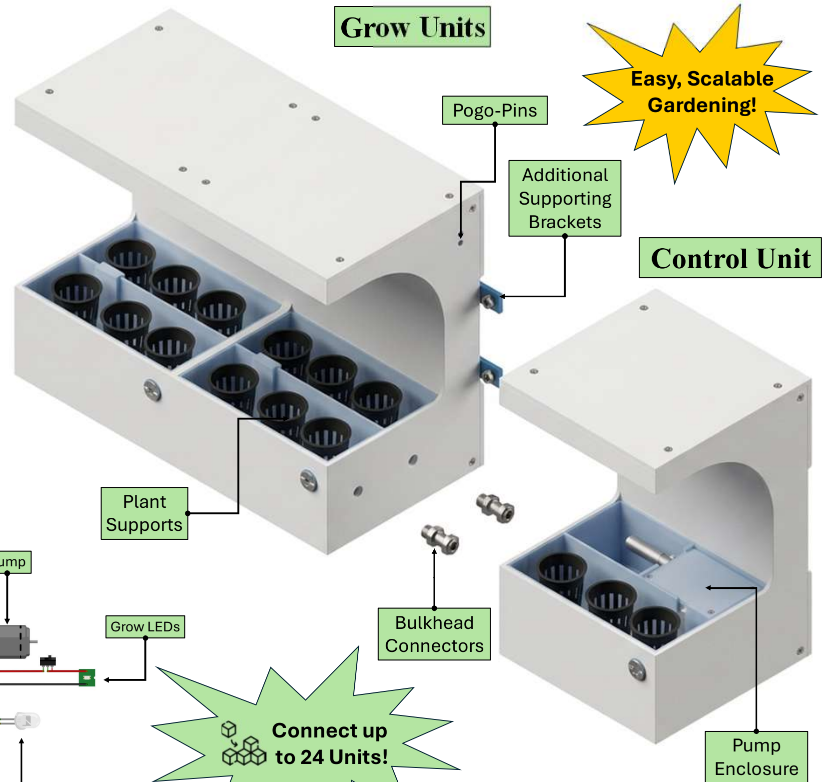
Pump Enclosure



Internal Circuitry



Grow Units



Easy, Scalable Gardening!

Connect up to 24 Units!

5. Testing & Results

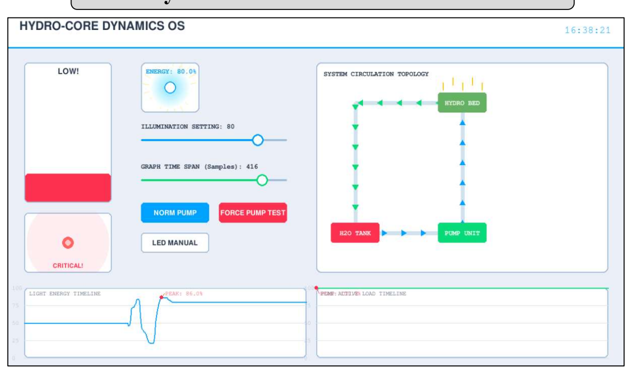
PASS — Leak Proof Assembly Test
- Ensure the integrity and leak resistance of assembly

PASS — Frame Strength Test - ASTM E8
- Verify stacked configuration load does not warp

PASS — Noise Test at 3ft - ISO 3744
- Confirm sound output does not exceed ≤ 40 dB

PASS — 72 Hour Reliability Test
- Verify system operation and automatic shut off/restart

System Monitor & Control Panel

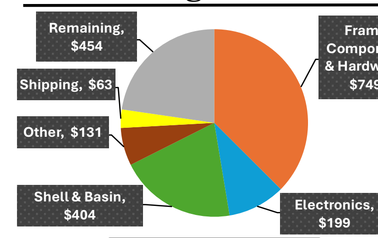


6. Why Gρ?

- Modularity means customizability!
- Simple assembly → Easy operation
- Hands-off automatic system minimizes user burden

Consumer price for 1 control unit and 2 grow units: $600

7. Budget & Work



Total Spent: $1,546

- **1,700 total hours of design, manufacturing, & testing**
- **60 Engineering Hours/Week**

8. Future Work

- Adopting mass manufacturing practices for scaled production
- Create an app and utilize internet capabilities for IoT integration
- Expand market reach toward hotels, office buildings, and the agricultural industry
- Improving modularity

9. Lessons Learned

- Create a persona for your user
- Effective communication reduces internal conflict
- Master modeling CAD techniques reduces time spent on modeling

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